

Current Trends in Artificial Intelligence

Project : MMA Fight Prediction

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1 Introduction

In this project, a Multi-Agent system was designed to predict the winner of MMA fights. Each agent acts as an expert, focusing on a specific task such as analyzing historical data, the news, the bookmakers' feelings or using Machine Learning for predictions. These agents' analyses are aggregated to guide an LLM, which is responsible for doing the final prediction. To improve transparency, the system provides explainable results using Diverse Counterfactual Explanations (DiCE). User can either interact through a chatbot or select a fight through a calendar.

2 Why we picked this topic ?

In today's world, teenagers are blinded by the will of being rich. One thing that reinforces this ideology is sports betting. Everywhere on the social media we can see fake bookmakers assuring they have found the perfect formula to find 95% or even 99% of the time the winner of a fight, a football match or anything else. By choosing to pursue this project, we aimed to critically assess the validity of these claims and determine whether these "miracle solutions" are grounded in data or are just predatory schemes targeting vulnerable audiences. In addition, this allows us to explore the topics of Multi-Agent systems, Explainable AI and LLMs seen during lectures. It is interesting to combine different subjects as it enables aggregating the strengths of each approach. We wanted to use a Multi-Agent system with an LLM as it allows mimicking some kind of critical thinking, where the LLM does not follow a single biased analysis but multiple evaluations. We think that Explainable AI is an unavoidable direction for the future of the field. In a context where AI systems' usage in decision-making increases, interpretability is crucial. Therefore, this project was the perfect opportunity to explore this topic we had never encountered in our academic journey.

3 Why we approached as we did ?

Since MMA fights are impacted by different factors, physical or psychological, we wanted to cover distinct angles. The Historical Agent observes head-to-head and win streaks to ground. The News Agent captures recent contexts such as injuries or mental states. The Bet Agent reflects the market sentiment about the fight by averaging the odds of experts. The Predictor Agent provides predictions using XGBoost. This model was chosen as it performs well in capturing complex, non-linear relationships and it is supported by the DiCE-ML library. We chose to use DiCE as it generated multiple counterfactual actions, which felt more informative than feature importance scores or a single action alone. We gave the LLM 3 scenarios of counterfactual actions, allowing it to have diverse perspectives for its prediction. These agents are coordinated by an Orchestrator as seen during lectures. This enables maintaining a structured workflow, controlling each agent output. We use *gpt-4o-mini* as it is a light model, with a large number of free tokens for students. To avoid exhausting API calls language processing and caching strategies were used. In the chatbot, the queries are first parsed using Spacy to proper nouns before any LLM call. Predictions are cached

for 24 hours and fight schedules for 7 days, as we estimate that these are unlikely to change within those delays. Regarding the data, we planned to scrape websites, but we were quickly banned from them, making our system not stable. For prediction, we rely on a fixed stats database updated 3 months ago. As fights are not that frequent for the fighter, the data remains approximately correct. We had to accept this trade-off, as we had no other reliable choice (no free API available).

4 What the main results are ?

Fighter 1	Fighter 2	Predicted	Pred. Method	Conf. (%)	Actual Result	Actual Method
Cameron Smooth	Kai Asakura	Cameron Smooth	Decision	51.3	Cameron Smooth	KO
Rei Tsuruya	Luis Gurule	Rei Tsuruya	Decision	59.7	Rei Tsuruya	Submission
Sergei Pavlovich	Tallison Teixeira	Sergei Pavlovich	KO/TKO	60	Sergei Pavlovich	KO
Jake Matthews	Carlston Harris	Jake Matthews	Decision	69.7	Jake Matthews	Decision
Jordan Leavitt	Joanderson Brito	Jordan Leavitt	Decision	64.6	Joanderson Brito	Submission
Priscila Cachoeira	Chelsea Chandler	Priscila Cachoeira	Decision	53.7	Chelsea Chandler	Submission
Bruno Silva	Edgar Chairez	Bruno Silva	Decision	69.5	Edgar Chairez	TKO
Brendan Allen	Edmen Shahbazyan	Edmen Shahbazyan	Decision	52.2	Edmen Shahbazyan	Decision
Fares Ziam	Tom Nolan	Fares Ziam	Decision	69.2	Tom Nolan	Decision
Matt Schnell	Alessandro Costa	Alessandro Costa	Decision	76.2	Alessandro Costa	TKO
Marcus McGhee	John Yannis	Marcus McGhee	Decision	63.6	Marcus McGhee	Decision
Belal Muhammad	Gabriel Bonfim	Gabriel Bonfim	Decision	71.1	Gabriel Bonfim	Decision

Table 1: Prediction results for May 30th (above) and June 6th (below) fight nights. Green rows are fully correct, yellow rows have the right winner but wrong method, red rows are incorrect predictions

Table 1 represents the results of the simulation on 2 fight nights. For the first, we see that our system performed really well. It successfully predicted 3 winners out of 4 fights, and successfully found the exact winning method for 2 of them. However, for the second night, we can observe a drop in performance. Our system made mistakes in 5 fights and only correctly predicted the winners of 3 fights, finding the exact winning method for 2 of them. Looking at the confidence scale, we can observe that the system tends to be more accurate when its confidence level is higher. But we can also notice that even with a high confidence score, it still occasionally makes mistakes. Ultimately, these results are a good reminder that MMA is a highly unpredictable sport. It is even more complicated when we are trying to predict the right winning method, which is sometimes hit or miss. A single mistake inside the octagon can lead to an unexpected KO or submission, which can cost the whole bet and makes the sport inherently very hard to predict.

5 What have we learned from the implementation ?

During this project, we have learned many things and some of them might be helpful for our future careers. First, it was the first time that we were using API or LLM calls in a project. We had this idea of a project that could meet the needs of the industry, instead of those abstract ones in other courses. It also extended our experience in terms of teamwork because we had a great group that split the work fairly, but also was there for each other in case anyone had an issue with their part of the job. For example, one of us was trying to do some scraping on different websites, but it kept being blocked by them. So, another one came with the idea of API, which fits perfectly with our idea of this project too. It showed us that a single problem can have many solutions. Before this project, we also never used any Machine Learning model, so we were thrown into the deep end, and we learned there about data leakage and feature extraction. It also helped us for other courses such as Statistical Foundation of Machine Learning. Lastly, we worked during this project with only free APIs, so we had to be ingenious about the calls we made. This constraint became a moment of creativity for us, as we tried to reduce the number of calls by extracting the names with Spacy (and if it fails then we use an LLM). Same idea for updating the fight calendar once every week to save some tokens, and lastly we've had a cache to keep the predictions for 24 hours if by any chance the user asks the same query on the same day. Hence, we learned a lot by doing this project.